STAT107_Halfway_Report

TEAM 5

2026-11-03

Abstract:

Give a title to your project. In an abstract of 200 words or less, describe what is the main question you aim to answer, and a brief summary of your current results.

- TITLE: Investigating Possible Income-Dependence in Disease and Illness
- This is an observational study using two publicly available data sets to analyze correlation between income and various health statuses. We investigate using median household income by county in the United States data from the US Census Bureau and population health data by county in the United States by the CDC. Using the computational tool R, we combine the data sets and isolate comparisons for analysis with the following findings. Health status seems to be negatively correlated with income. That is, the data shows counties with higher median household incomes had less adverse health statuses across various types of health status and various income brackets. The report will provide two analyses delving into income's relation to diabetes and depression and provide visual maps which depict the correlation we found.

Intro

What is the central question that motivates your analysis? Who may this analysis benefit? Are you trying to train a model, test a hypothesis, classify objects apart? Why? This has to be written with proper grammar and syntax: sentences, paragraphs, and a narrative structure.

- The question we are seeking to answer is whether income (whether it be higher or lower) has an effect on disease prevalence. Will median household income show a correlation to disease proclivity? If an increase in income does prove to have a negative affect on disease prevalence, the information we find could possibly help find a solution to this issue for interested parties. This can manifest in providing easier access to necessary healthcare for lower-income individuals, or helping better conditions to prevent the diseases from occurring. Alternatively, certain companies and investors may benefit from such a relationship being found by providing insight into the likelihood of certain investments succeeding in a population due to their proximity to disease and their relative wealth.

To realize these possible benefits, we are creating correlation maps using publicly available data on the incomes of household in the US and comparing these numbers to CDC estimates from 2025 for health status.

Executive Summary

- Provide, in a series of 3-5 bullet points, what are the core conclusions and findings of your analysis so far.

1. Negative correlation exists between increasing income and disease rate. We are projecting our observations of diabetes and depression prevalence in correlation to income for this conclusion, but further analysis of different health statuses is possible.
2. Through a regression model for diabetes, we see that Median Household Income is statistically significant, and that it accurately explains ~45% of the variation in diabetes prevalence.
3. Through linear and quadratic regression we can see that the model for diabetes is quadratic, with diabetes prevalence decreasing at lower incomes, but slowly leveling out. At some point in the higher incomes, the line even upturns very little. Minimum diabetes prevalence is around median income of $108k.
4. A graph for depression prevalence in correlation to income shows a similar relationship. It is also a quadratic line representing a negative relationship with a decreasing prevalence as income increases. However, the most notable difference is the depression graph doesn't show any upturn at any moment, suggesting that prevalence continually decreases. Depression rates drop continually as income increases.

Data

What is the data available for performing the analysis? Where did you retrieve these data from? What are the variables and number of observations available? What steps where performed to clean the data? If the data is obtained via randomized simulations, clarify the data generating procedure

- US Census Bureau's data set for "Median Household Income in the Past 12 Months in 2023" @ https://data.census.gov/table/ACSDT5Y2023.B19013 This data set includes 3222 observations across 5 variables representing the geographic ID of a county, the name of the county, the median income, the margin of error, and the FIPS codes of the households surveyed. We used this data set to show the different median household income in each US county. To clean the data set, we first renamed the columns for clarity. Then, we changed the GeoId into a compatible FIPS code.
- CDC data set for "Local Data for Better Health, County Data in 2025" @ https://data.cdc.gov/500-Cities-Places/PLACES-Local-Data-for-Better-Health-County-Data-20/swc5-untb/about_data This data set includes 229298 observations across 22 variables representing both location names (geographic ID, state name, etc) and the estimates for measures (health outcomes, health status, disability, etc) within those locations. Using this data set, we can find the percentage amounts of disease from each county. To clean this code, all we did was convert the FIPS code into a numeric.

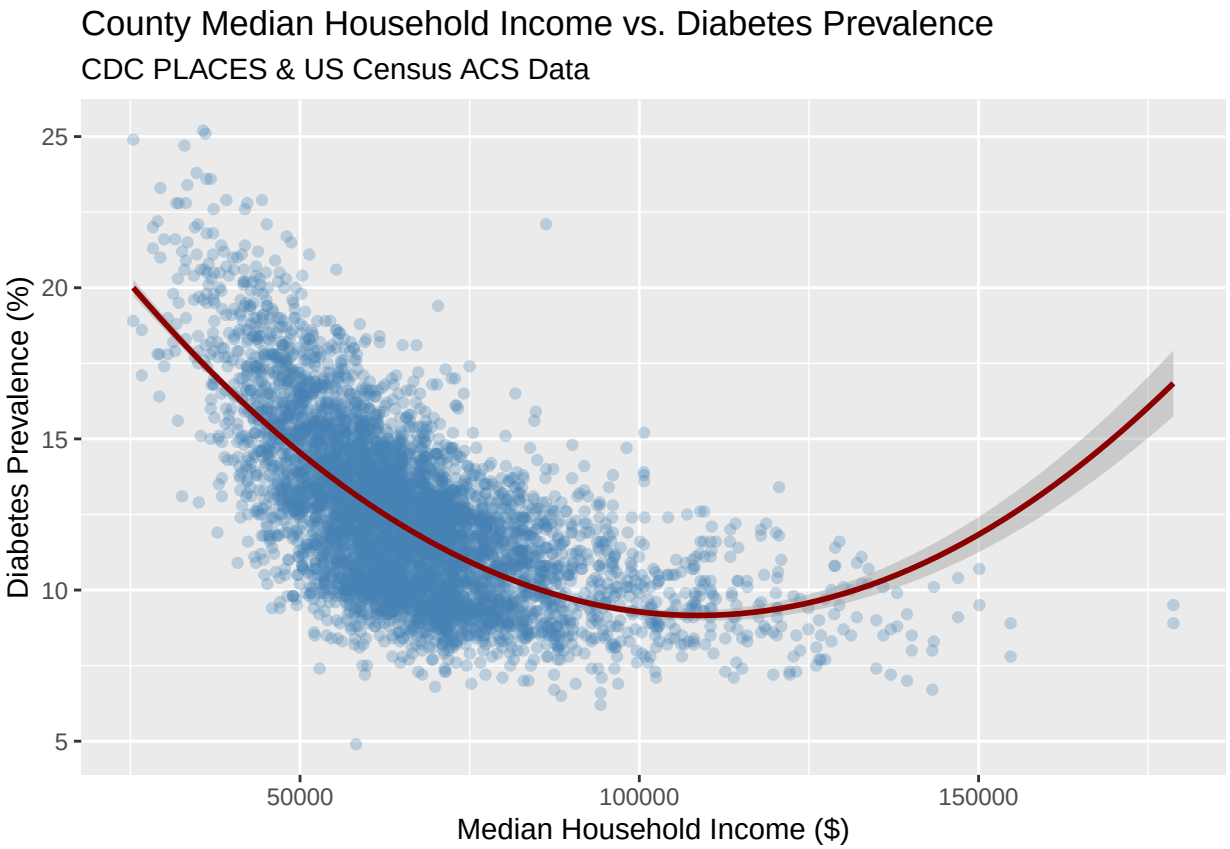
  With the FIPS code from each data set, we were able to merge the data sets and compare them directly. To clean this new data set, we changes the Median Household Income and the Total Population columns into numerical data. Lastly, we kept only the necessary columns that we were going to use.

  After all of the data cleaning was done, we saved the new data set into a new .csv file.

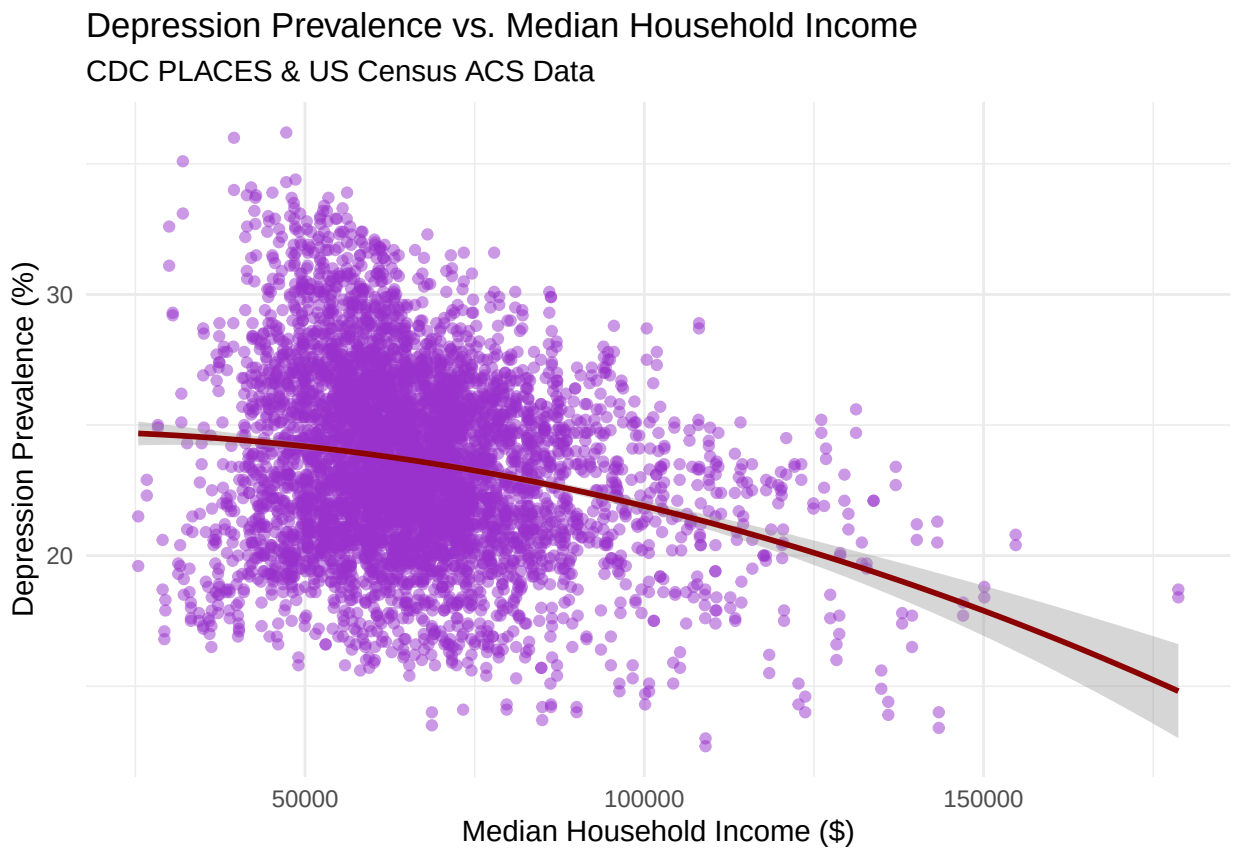
Visualization

Provide preliminary visualizations in the form of histograms, density plots, scatter plots, box-plots, and numerical summaries of the data, depending on the type of analysis. What do these visualizations suggest about the hypothesis, model or simulation you are aiming to perform? If you aim to generate further data visualizations, describe them here.

- By utilizing a simple scatter plot and a regression line, we can see that diabetes has a negative relationship with Median Household Income, meaning that as income increases diabetes is less prevalent. However, seeing that the line is quadratic, we can see minimum diabetes levels minimize at around $108k median income. At every income over $108k, prevalence starts to rise again.



- We used a similar method to visualize depression percentages. Similar to the diabetes graph, it is quadratic. However this graph is different as it is a negative quadratic, so as income continues to increase, prevalence for depression decreases. Interestingly, at higher incomes, the decrease is slighter; the change becomes flatter.



Analysis

Provide some preliminary plans of what analysis you aim to complete. Mention models, tests, procedures, and some ideas of what expected outcomes you hope to find

- So far, our project covers the models of 2 diseases in relation to household income. Later on, we plan on creating more models, possibly around 5 (of all different types– heart conditions, strokes, obesity, etc.) We aim to do this in order to get more knowledge on the unknown advantages of having higher income. Data will be compared and analyzed thoroughly, and we will see if each disease continues to hold a negative relationship against median income or if there are some that have positive correlation. In the end, we hope to find at least one disease that has a positive correlation. We strive to find logical reasoning as to why this phenomena takes place– is it simply access to better healthcare? is it inability to spend money on better food/accomodations? is it a difference in way of life and expectations between different income brackets? By comparing data, reading literature, and analyzing our results we intend on creating some sort of answer.

Summary of Individual Contribution

Vanessa: Added to the 01_DataCleaning.R file: Renamed columns for legibility, removed first row of raw data (descriptive text), converted necessary variables to numerical values (R was reading them as text instead of numbers), created a new .csv file for the merged and clean data

Created the 02_DiabetesAnalysis.R file: Creating a dataframe for Diabetes disease, summarize results from Diabetes dataframe (max, min, standard deviation for diabetes prevalence & household income), scatterpolt with regression line for visualization that compares median household income with diabetes prevalence

Created the FinalReportDraft and added the appropriate sections required for the report including the Summary of Individual Contributions

Contributed to README file describing the raw data sets, and the 00, 01, and 02 r files.

Alexandra: Created the 00_requirements.R file: added all necessary packages.

Created the 01_DataCleaning.R file: Found, interpreted, called, cleaned, and combined the data.

Added to the 02_DiabetesAnalysis.R file: Combed through data, corrected models. Initial models showed linear relationship, through more analysis (regression models) quadratic is a better fit.

Finalized FinalReportDraft.Rmd: Fixed the code on the plot portion, removed unnecessary info, changed bullet-pointed information into finalized paragraphs.

Created 10_Cleaning&Processing.RMD file: wrote explanations, just transferred code.

Created 11_Visualizations.RMD file: wrote explanations again, just transferred code.

Created and shared GitHub Repository

Andrea: Edited FinalReportDraft.Rmd (the report, or report) into an outlined draft report. Completed ABSTRACT, INTRO, and DATA drafts. Outlined Analysis, Visualization, and Executive Summary drafts.

Initially edited the report to include data cleaning (renamed variables, removed unneeded data) for ease of access. (History @ https://github.com/ahemm005/Stats-107-Final-Project/commit/8d2bf25e9903be5215166167e943f4282743f62f)

Created depression analysis with code in the report.

Proofread and edited finalized report for readability and clarity of purpose.

Created README guide for reproducability.